WiLiMan-ID

Funded by the European Union

HPAI Modelling Challenge

HPAI epidemic narrative – Phase 2

Summary of interventions implemented since 13 January 2026

It has now been 53 days since the virus was first detected on the island (22 December 2025), and the epidemic is still not under control (see below). Since the Phase 1 data cut-off (13 January 2026), several management strategies that had been implemented earlier were maintained. They include:

- **Reactive culling**: same approach as in Phase 1; restocking of the depopulated farms (whether they had been reactively or preventively depopulated) is still not allowed.
- **Contact tracing**: same approach than in Phase 1
- **Zoning**: same approach than in Phase 1
- **Movement bans in the regulated zones**: same approach than in Phase 1
- **Enhanced biosecurity and detectability in regulated zones**: same approach than in Phase 1
- **Pre-movement testing in the HRZ**: same approach than in Phase 1

However, in an effort to accelerate epidemic control during this recent period, some strategies were adapted and others were introduced. They include:

- **Preventive culling:** to avoid saturating the culling teams, and given the apparent important role chicken flocks play in virus dissemination, blanket preventive culling of all flocks within 1 km of infected premises was discontinued. Instead, the preventive culling radius was extended to align with that of the protection zones (within 3 km of infected premises) and was focused on chicken flocks (layers and broilers) only.
- **Confinement**: as stage 2 broiler farms and organic duck farms typically allow poultry to have outdoor access—leading to potential contact with contaminated environment or contributing to environmental contamination—a temporary confinement regulation was imposed for these farm types. It is expected that the likelihood of these farms becoming infected or spreading the infection is reduced.

Note that, thanks to collaborations with companies from neighbouring islands, culling capacity has been substantially improved through a rapid mobilisation effort over several weeks. Additional teams were deployed in successive waves, increasing the national daily culling limit to several dozen farms per day by the end of the ramp-up period. The culling capacity is no longer saturated.

Summary of the epidemiological situation between 13 January and 13 February 2026

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Since the Phase 1 data cut-off, 363 outbreaks were reported, corresponding to a 3-fold increase as compared to the first few weeks of the epidemic.

At the beginning of this interval, daily outbreak incidence continued to increase erratically, with a few long-distance translocation events suspected to be linked to contact with infected wildlife or long-distance shipments of latently infected stage 1 broiler flocks. As a consequence, the virus has now been reported in most regions of the island, with many outbreaks reported in the two regions of high poultry farm density.

During the latter part of this epidemic period, the observed daily incidence began to decline, indicating an improvement in the situation. However, several infection foci remain active and transmission has likely not been fully interrupted. Fortunately, in the past few days, daily case notifications have fallen to single-digit levels, though sporadic outbreaks continue to be reported.

Data provided

Attached to this narrative, you will find a series of files containing all epidemic data collated by the veterinary services since the start of the epidemic (including data provided in Phase 1). It is expected that these sets of data are still accurate. They include:

- **cases**: same format as in Phase 1.
- **preventive culls**: same format as in Phase 1.
- **movements**: same format as in Phase 1.
- **activity**: same format as in Phase 1.

Your mission is to exploit these data and send us a report that addresses the following items:

1. A general description of the ongoing epidemic, including a table describing the distribution of the outbreaks per species and production types, a timeline of the incidence and a visual representation of the spatial distribution of the outbreaks.
2. A prediction of the likely temporal and spatial evolution of the epidemic over the following four weeks, assuming the management strategy remains as it is today; if you can have it by farm type, that would be nice.
3. An answer with a justification to the following question: given that outbreak farms are likely more infectious than preventively-culled farms, how epidemiologically-relevant would it be to ignore the preventive culling actions of chicken farms and spend all efforts possible to reduce by 1 day the start of the reactive culling actions in outbreak farms?
4. An answer with a justification to the following question: given that duck farms likely to contribute less to the epidemic than chicken farms, what would be the epidemiological consequences of stopping the confinement of the organic ducks? And of stopping the confinement at all, even in chicken farms?
5. An answer with a justification to the following question: how epidemiologically-relevant would it be to prohibit restocking of all farms (not just those that were depopulated) located within declared surveillance zones (e.g. 10 km around detected outbreaks)?

Your report is due on 31 March 2026.

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We acknowledge that we ask a lot, and that not all modelling approaches can address all the questions. Questions 1 and 2 should be addressed in priority. An investigation of questions 3 4 and 5 would be an immense bonus for us.

Reporting outputs

In addition to a narrative report (in .pdf) that addresses the questions above, please update the model description template you submitted at the end of phase 1 if your modelling approach has been modified.

For any temporal prediction, please share raw simulated trajectories as a .csv file with an accompanying description (e.g. README file). If raw trajectories are not available, please share synthetic metrics of the trajectories at different time points (at least at the end of the four weeks prediction), if possible with distributions (preferably) or at least with confidence/credible intervals, when relevant.

For spatial predictions, please share the spatial risk of the spatial units of interest at specific time points (at least at the end of the four weeks prediction) as a .csv file (preferably) or a .shp file. As for temporal predictions, please provide a description of the file.

Finally, we would appreciate if you could draft a one-page (max!) recommendation brief synthesising your key findings and their implications in terms of epidemic management.

Good luck everyone!

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